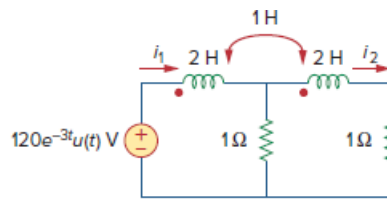


Problem

Find $I_1(s)$ and $I_2(s)$ in the circuit of Fig. 16.92.

Figure 16.92:



Step-by-step solution

Step 1 of 9

Refer to the Figure 16.92 in the textbook.

Consider the source voltage,

$$v(t) = 120e^{-3t}u(t)$$

In Laplace domain,

$$V(s) = \frac{120}{s+3}$$

The impedance offered by the inductors is,

$$\begin{aligned} X_L &= sL \\ &= 2s \end{aligned}$$

Step 2 of 9

Sketch the circuit in Laplace domain as shown in Figure 1.

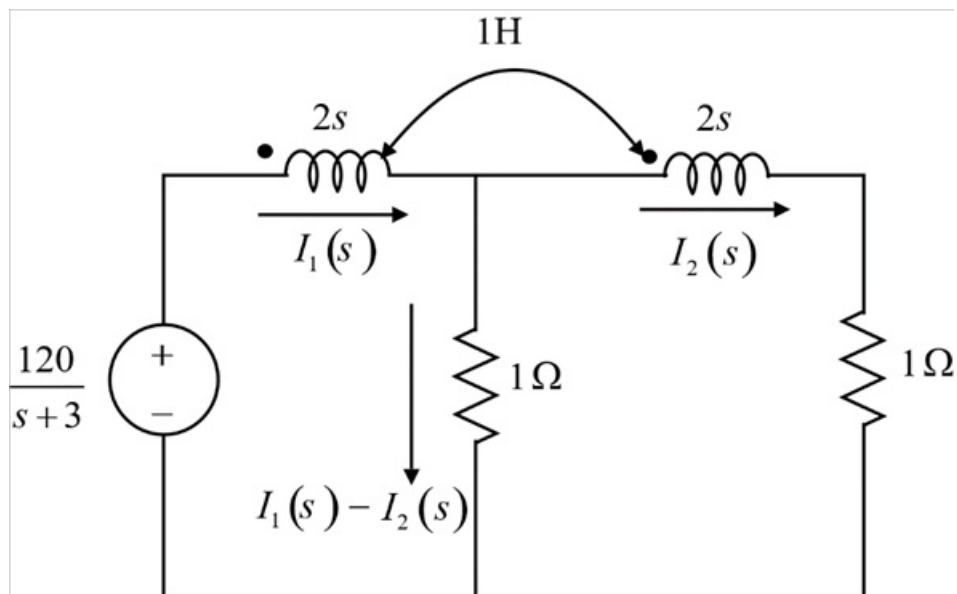


Figure 1

Step 3 of 9

Since the current $I_2(s)$ is entering the dot of second inductor, the polarity of the mutual voltage at the dot of the first inductor is positive.

The mutual voltage is,

$$v_{m1} = M \frac{di_2}{dt}$$

$$V_{m1}(s) = sMI_2$$

$$V_{m1}(s) = sI_2$$

Thus, replace the effect of the mutual inductance with a voltage source of $sI_2(s)$ V keeping in mind the polarity as shown in Figure 2.

Step 4 of 9

Since the current $I_1(s)$ is entering the dot of first inductor, the polarity of the mutual voltage at the dot of the second inductor is positive.

The mutual voltage is,

$$v_{m2} = M \frac{di_1}{dt}$$

$$V_{m2}(s) = sMI_1$$

$$V_{m2}(s) = sI_1$$

Thus, replace the effect of the mutual inductance with a voltage source of $sI_1(s)$ V keeping in mind the polarity as shown in Figure 2.

Step 5 of 9

Sketch the modified circuit as shown in Figure 2.

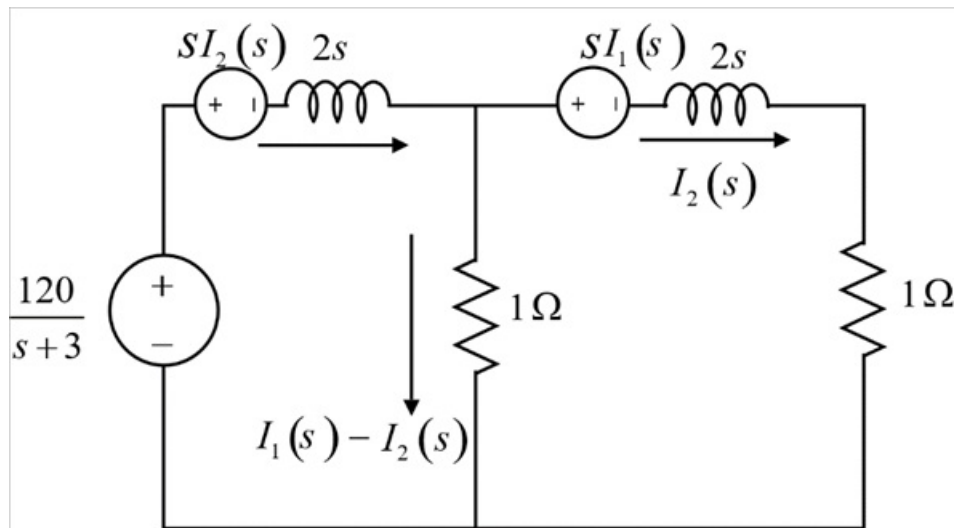


Figure 2

Step 6 of 9

Apply Kirchhoff's voltage law to the loop 1.

$$\frac{120}{s+3} - sI_2(s) - 2sI_1(s) - 1(I_1(s) - I_2(s)) = 0$$

$$\frac{120}{s+3} = (1+2s)I_1(s) + (s-1)I_2(s)$$

$$(2s+1)I_1(s) + (s-1)I_2(s) = \frac{120}{s+3} \dots\dots (1)$$

Step 7 of 9

Apply Kirchhoff's voltage law to the loop 2.

$$(1)(I_2(s) - I_1(s)) + sI_1(s) + 2sI_2(s) + I_2(s) = 0$$

$$(s-1)I_1(s) + (2s+2)I_2(s) = 0$$

$$I_2(s) = -\left(\frac{s-1}{2s+2}\right)I_1(s) \dots\dots (2)$$

Step 8 of 9

Substitute $-\left(\frac{s-1}{2s+2}\right)I_1(s)$ for $I_2(s)$ in equation (1).

$$(2s+1)I_1(s) + (s-1)\left(-\left(\frac{s-1}{2s+2}\right)I_1(s)\right) = \frac{120}{s+3}$$

$$I_1(s) \left[2s+1 - \frac{(s-1)^2}{2s+2} \right] = \frac{120}{s+3}$$

$$I_1(s) \left[\frac{4s^2 + 2s + 4s + 2 - (s^2 - 2s + 1)}{2s+2} \right] = \frac{120}{s+3}$$

$$I_1(s) \left[\frac{3s^2 + 8s + 1}{2s+2} \right] = \frac{120}{s+3}$$

Thus,

$$\begin{aligned} I_1(s) &= \frac{120(2s+2)}{(s+3)(3s^2+8s+1)} \\ &= \frac{240(s+1)}{(s+3)(3s^2+8s+1)} \text{ A} \end{aligned}$$

Therefore, the value of the current, $I_1(s)$ is $\boxed{\frac{240(s+1)}{(s+3)(3s^2+8s+1)}}$.

Step 9 of 9

Substitute $\frac{240(s+1)}{(s+3)(3s^2+8s+1)}$ for $I_1(s)$ in (2).

$$\begin{aligned} I_2(s) &= -\left(\frac{s-1}{2s+2}\right)\left(\frac{240(s+1)}{(s+3)(3s^2+8s+1)}\right) \\ &= -\frac{1}{2}\left(\frac{s-1}{s+1}\right)\left(\frac{240(s+1)}{(s+3)(3s^2+8s+1)}\right) \\ &= -\frac{120(s-1)}{(s+3)(3s^2+8s+1)} \end{aligned}$$

Therefore, the value of the current, $I_2(s)$ is $\boxed{-\frac{120(s-1)}{(s+3)(3s^2+8s+1)}}$.